A 400 MW offshore wind farm connection:

load flow, cable sizing, fault levels, protection grading and N-1

A self-directed study in four blocks (LV12 to LV15): the final base design, revision D

Jack Davies, BEng (Hons) Renewable Energy Engineering · 28 October 2026 · decisions D1 to D4 approved by the author; D5 to D7 taken on his instruction

Read this first. This is a self-directed study on assumed data. No real wind farm, cable datasheet, site survey or network operator data stands behind it: every input is either a design choice or a typical-order value written from general knowledge, and is marked ASSUMED. No standard (IEC 60287, IEC 60909, IEC 60255, the Grid Code or the SQSS) was read; the methods are written in the form those standards use, from general knowledge, and nothing here claims compliance with any of them. It is not professional experience and has not been reviewed by a chartered engineer. What it does show is the method: criteria written before any number was run, every result produced by code that reruns from a clean shell, independent checks, and every failure reported rather than hidden.

Summary

The connection is two 60 km, 220 kV export circuits from an offshore substation to a 400 kV point of connection, with two 250 MVA transformers at each end and five 66 kV strings of eight 10 MW turbines. The final base design, revision D, has two 155 MVAr onshore reactors, a 630 mm2 export cable and a 1000 mm2 array cable sized on a conservative soil value, and the string relay set at 960 A. Intact, all four operating cases are inside limits.

It is not N-1 secure at full output, which is expected of a design with two of everything at half rating. Losing one export circuit, one offshore transformer or one onshore transformer forces curtailment to about 240 to 250 MW at unity power factor (C1: 246.1, 248.4 and 239.8 MW). Losing a string's first cable section loses only that string and rides through in every case. Losing one of the two reactor units rides through at full output and leaves only the light-load case over voltage (PCC 1.0651 pu), where in revision C, with one reactor, the same outage put every case out of limits at full output.

Fault levels hardly move on the final cables (largest change 0.266 percent), so the grading of LV14 stands, with the onshore transformer relay at TMS 0.35 for discrimination at the minimum fault (backup time about 3.9 s at the maximum offshore fault, a finding) and the string relay at 960 A (0.4435 s ahead of the next relay). Every input is assumed; section 12 says what the study is not.

1. Question, scope and how the study was run

The study answers one question per block. LV12 built and checked an independent load-flow solver. LV13 asked whether four operating cases of the connection solve inside limits written in advance. LV14 asked what cable sizes, fault levels and overcurrent settings the design needs. LV15, this block, asks: which single outages can the connection ride through at full output inside limits, and for each that cannot, what is the largest output that stays inside limits?

Each block wrote its acceptance criteria into a file and committed it before the first number was run (LV15: model/criteria_C.json, commit 34caf6a, and model/criteria_D.json, commit df09ba4). A result outside a limit is an exception, explained by a computed cause; a remedy appears only as a separate labelled what-if, chosen after the result was seen, and the base case is never changed to make it pass.

Decisions

- D1. Sizing margin: cables are sized on soil thermal resistivity 1.5 K m/W (the LV14 sensitivity value) instead of 1.0, until a survey replaces it. This gives 800 mm2 array and 630 mm2 export from LV14's own sensitivity table. Proposed 27 Oct 2026; approved by Jack, 28 Oct 2026.
- D2. LV13 revision C: yes. The load flow uses the sized cables' resistance at 90 C and capacitance from LV14's cable model; reactance and lengths unchanged from LV13 (A07, A11). Ratings are those at soil 1.5 K m/W.
- D3. The minimum-fault ordering check joins the grading pass rule; R4 TMS 0.35 (LV14 what-if). The 3.9 s backup time this gives at the maximum offshore 66 kV fault is reported as a finding, not accepted silently.
- D4. No datasheets or survey exist: every input stays ASSUMED and the report says so in its first paragraph.
- D5. The array is re-sized on the revision C design current: 1000 mm2 (what-if W3 of revision C; the sizing settles there on soil 1.5 K m/W).
- D6. The onshore reactor is two 155 MVAr units (what-if W4 of revision C), so one outage leaves half the compensation. Outage O5 becomes one unit out.
- D7. R1 pickup 960 A (at least 1.2 times the revision C array load, 955.2 A).

Approved by Jack, 28 Oct 2026, 02:14 ('Approve D1 to D4 if it makes improvements'): D1 sizes on the conservative soil value, D2 uses the sized cables, D3 adds the minimum-fault check, D4 keeps every input honest; each makes the design or the study more robust. Taken on Jack's instruction of 28 Oct 2026 to finalise the studies with the improvements; each was shown first as a labelled what-if or finding on revision C. Reversible: revision C stays in the record.

2. The network

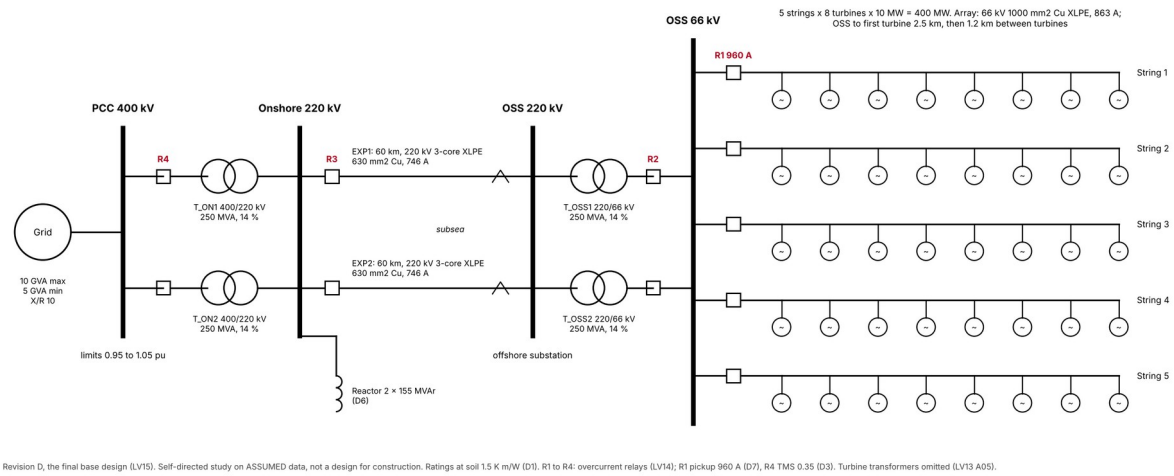


Figure 1. Single-line diagram of the final base design, revision D (results/sld.png). R1 to R4 are the overcurrent relays graded in LV14.

The model is a single neutral table read by two solvers: pandapower 3.5.6 and the independent Newton-Raphson solver from LV12. The remote grid is a fixed voltage behind a Thevenin impedance of 10 GVA at X/R 10; turbines are constant P and Q injections at their 66 kV array buses (turbine transformers are omitted, so turbine terminal voltage is not assessed); transformer taps are fixed at nominal.

3. Assumptions

Every input of the model. DESIGN is the author's choice; ASSUMED is a typical-order value, NOT SOURCED to a datasheet or standard. Revision D replaces the cable rows A07, A08, A11 and A12 with the values below, and A14 with two 155 MVAr reactor units in every case.

ID	Assumption (LV13)	Value	Unit	Status
A01	Installed wind capacity	400.0	MW	DESIGN
A02	Turbines per string	8	-	DESIGN
A03	Strings	5	-	DESIGN
A04	Turbine rating	10.0	MW	DESIGN
A05	Turbine transformers (0.69/66 kV) and turbine terminals	omitted	-	DESIGN
A06	Array cable section lengths: OSS to first turbine / turbine to turbine	2.5, 1.2	km	ASSUMED
A07	Array cable (66 kV, one type for all sections) R / X / C	0.04, 0.1, 200.0	ohm/km, ohm/km, nF/km	ASSUMED
A08	Array cable provisional continuous rating	800.0	A	ASSUMED
A09	OSS transformers 66/220 kV: number, rating, impedance voltage, resistive part	2, 250.0, 14.0, 0.3	-, MVA each, %, %	ASSUMED
A10	Export cable: circuits, length	2, 60.0	-, km	DESIGN
A11	Export cable R / X / C	0.022, 0.125, 170.0	ohm/km, ohm/km, nF/km	ASSUMED
A12	Export cable provisional continuous rating per circuit	1000.0	A	ASSUMED
A13	Onshore transformers 220/400 kV: number, rating, impedance voltage, resistive part	2, 250.0, 14.0, 0.3	-, MVA each, %, %	ASSUMED
A14	Onshore shunt reactor (220 kV), sized as 50 percent of total export cable charging at 220 kV, r	computed	MVAr	DESIGN
A15	Grid at the 400 kV point of connection: short-circuit level, X/R	10000.0, 10.0	MVA, -	ASSUMED
A16	Frequency	50.0	Hz	DESIGN
A17	Shunt conductance and cable dielectric loss	omitted	-	DESIGN
A18	Power factor convention	0.95	-	DESIGN
A19	Reactive limits and OLTC action	omitted	-	DESIGN

Revision D cable	Size	R at 90 C (ohm/km)	C (nF/km)	Rating at soil 1.5 (A)	Design current (A)	Margin
66 kV array (sized on rev C current)	1000 mm2	0.0284	323.3	863.3	796.0	8.5 %
220 kV export	630 mm2	0.0379	144.1	745.9	738.9	1.0 %

Cable inputs (LV14, all ASSUMED): copper conductor at 90 C, seabed 15 C, buried 1.0 m, XLPE 10 mm (66 kV) and 24 mm (220 kV) at 3.5 K m/W, coverings 16 mm at 6.0 K m/W, sheath and armour loss 10 percent of conductor loss. Soil thermal resistivity 1.5 K m/W in revisions C and D (D1), against 1.0 in LV14. Reactance is unchanged from LV13.

4. Acceptance criteria

- Voltage: PCC 0.95 to 1.05 pu; every 220 kV and 66 kV bus 0.90 to 1.10 pu. Author's criteria, NOT SOURCED to the Grid Code.
- Loading: cables as current over the rating at soil 1.5 K m/W; transformers as apparent power over nameplate; each at most 100 percent.

- N-1 rule: run each outage at the case's output; if inside limits it RIDES THROUGH; if not, find by bisection to 1 MW the largest output, all in-service turbines scaled equally, that is inside limits (CURTAILED). A run that does not converge counts as outside limits.
- Grading (LV14 plus D3): pickup at least 1.2 times maximum load and at most 0.8 times the minimum fault at the zone end; 0.30 s between successive relays at the maximum and, from revision C, the minimum fault.

5. Load flow, intact network

LV13 (revision A, provisional cables, reactor in C4 only) found every case outside limits, with one cause: the two 60 km export circuits generate about 310 MVAr of charging power that lifts every voltage behind them and loads the onshore transformers to 108 and 126 percent in C1 and C2. Revision B added 310 MVAr of reactor in every case (decision 7, 21 October), which brought C1 to C3 inside limits and left C4's PCC at 1.0521 pu. Revision C added the cables sized on soil 1.5 K m/W; revision D, below, is the final base design.

Case	Gen (MW)	Loss (MW)	PCC (pu)	V max (pu)	V min (pu)	Export (A)	Array 1st (A)	Max cable %	Max trafo %	Verdict
C1	400.0	9.00	0.9902	1.0000	0.9627	606.6	712.2	82.5	83.5	inside limits
C2	400.0	9.96	0.9537	1.0377	0.9500	706.7	725.9	94.7	83.0	inside limits
C3	400.0	10.35	1.0246	1.0500	0.9175	593.7	794.1	92.0	95.4	inside limits
C4	40.0	0.72	1.0481	1.0641	1.0421	398.1	79.6	53.4	9.1	inside limits

Cases: C1 400 MW unity power factor, grid 1.00 pu; C2 400 MW exporting reactive power at 0.95, grid 0.95 pu; C3 400 MW absorbing reactive power at 0.95, grid 1.05 pu; C4 40 MW unity, grid 1.05 pu. Source: results/n1_D.json.

Why revision D: on revision C the C3 case loaded the first section of every string to 100.73 percent. The 800 mm2 cable chosen for the conservative soil has more capacitance than the 630 mm2 cable whose current it was sized from, so its own charging current pushed the current (796.0 A) above its rating. Sizing has to be iterated until it settles; one more step, to 1000 mm2 (863.3 A), clears it (D5), at 92.0 percent. The C4 PCC voltage, 1.0521 pu in revision B, is 1.0481 pu now, inside the limit, because the 630 mm2 export cable has less capacitance than LV13 assumed.

6. Cable sizing (LV14, then revisions C and D)

Method: a one-dimensional thermal circuit of the form IEC 60287 uses. Heat from each core crosses its own insulation (T1), then the heat of all three cores crosses the coverings (T3) and the soil (T4); the rating is the current at which the conductor reaches 90 C. LV14 chose 630 mm2 (array, 814.0 A against 785.7 A) and 500 mm2 (export, 759.6 A against 738.9 A) on soil 1.0 K m/W, margins of 3.6 and 2.8 percent. Its own sensitivity showed both fail at soil 1.5 K m/W, the least certain input, so the final design sizes on 1.5 (D1): 630 mm2 export, and the array settles at 1000 mm2 once its own charging current is included (D5).

Worked example, 500 mm2 export cable (LV14 results/worked_example.txt), re-derived independently on 27 October 2026 (LV14/independent_check/check.py, all twelve values within 0.12 percent):

1 Temperature rise available: 90.0 - 15.0 = 75.0 K

2 Conductor diameter from area: sqrt(4 x 500 / (pi x 0.92)) = 26.31 mm

3 AC resistance at 90 C: Rdc20 = 1.7241e-8 / 500e-6 = 34.48 uohm/m; x(1+0.00393 x 70) = 43.968 uohm/m; x(1 + ys 0.0411 + yp 0.0134) = 46.367 uohm/m

4 Insulation: T1 = 3.5/(2 pi) x ln(1 + 2 x 24 / 26.31) = 0.5784 K m/W

5 Coverings: outer diameter 192.1 mm; T3 = 6.0/(2 pi) x ln(Dext / Dpre) = 0.1740 K m/W

6 Soil: u = 2 x 1.0 / 0.1921 = 10.411; T4 = 1.0/(2 pi) x ln(u + sqrt(u^2 - 1)) = 0.4828 K m/W

7 Dielectric loss per core: Wd = w C U0^2 tan d, C = 133.94 nF/km, = 0.6789 W/m

8 Numerator: 75.0 - 0.5 x 0.6789 x 0.5784 - 3 x 0.6789 x 0.6568 = 73.4660

9 Denominator: R T1 + 3 R (1 + 0.1) (T3 + T4) = 1.2732e-04

10 I = sqrt(73.4660 / 1.2732e-04) = 759.6 A (cable.py: 759.6 A)

N-1 consequence: if one export circuit is lost, the other has to carry about twice the current. No size up to 2000 mm2 rates for that on LV14's inputs, so N-1 security on the export would need a third circuit or a much larger single cable, not a larger cable of this type. This is why section 9 finds curtailment, not ride-through.

7. Fault levels

IEC 60909 voltage-source method as coded in pandapower (the transformer correction factor and c factors were read in its source; the standard was not read). Maximum: three-phase, c 1.10, grid 10 GVA, all in service, wind at 1.1 times rated current per turbine. Minimum: two-phase, c 1.00, grid 5 GVA, no wind.

Busbar	kV	Ik'' max (kA)	grid only (kA)	ip (kA)	Ik'' min 2ph (kA)	Ik'' max rev D (kA)	Change
PCC	400	15.07	14.43	36.5	6.25	15.07	+0.00 %
ON220	220	8.76	7.60	21.8	4.74	8.76	+0.00 %
OSS220	220	7.46	6.31	17.9	4.07	7.45	-0.14 %
OSS66	66	17.07	13.23	40.3	9.04	17.06	-0.06 %
S1T8	66	12.72	9.76	27.2	6.91	12.75	+0.27 %

Hand check at the offshore 66 kV bus (grid only, maximum), on 100 MVA: grid 0.00109 + j0.01095 pu, two onshore transformers in parallel 0.000578 + j0.02699 pu (K_T 0.964), two export circuits in parallel 0.001364 + j0.007748 pu, two offshore transformers 0.000578 + j0.02699 pu. Total |Z| = 0.07276 pu; Ik'' = 1.1 x 0.8748 / 0.07276 = 13.23 kA, against 13.226 kA in the model (independent check, 0.004 percent). Single-phase earth faults are not studied: no zero-sequence or earthing data has been assumed.

8. Overcurrent grading

Four relays on the IEC standard-inverse curve t = TMS x 0.14 / ((I/Is)^0.02 - 1) (curve constants seen in relay-vendor summaries, NOT VERIFIED). R1 string feeder (66 kV, Is 950 A in LV14, 960 A from D7, TMS 0.10), R2 offshore transformer incomer (66 kV, 2400 A, 0.10), R3 export circuit onshore end (220 kV, 900 A, 0.125), R4 onshore transformer HV side (400 kV, 400 A, 0.225 in LV14, 0.35 from D3).

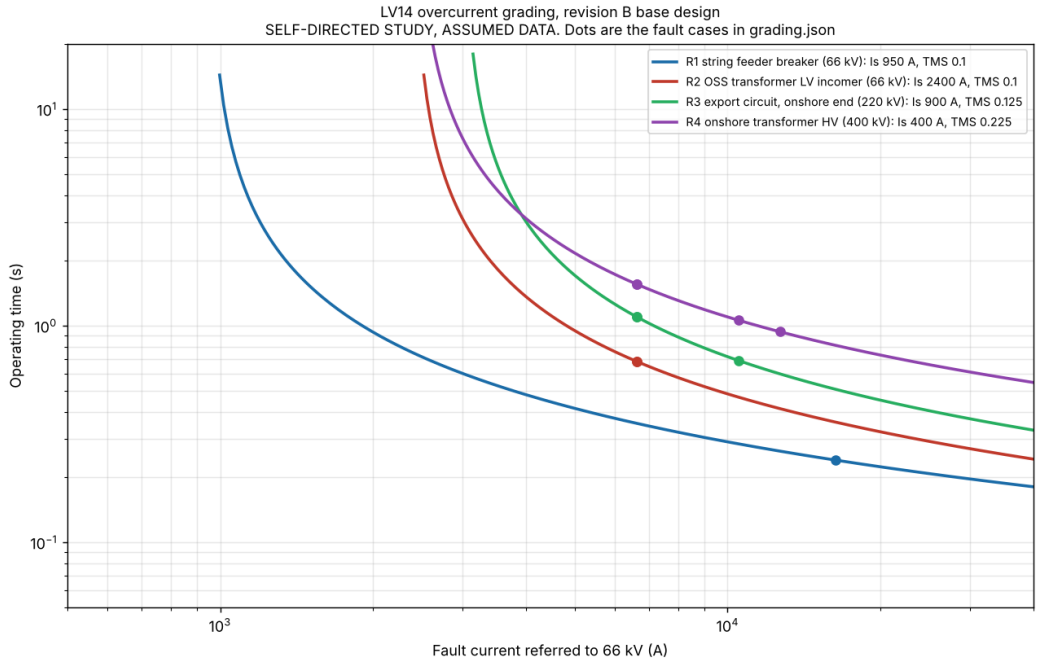


Figure 2. Grading chart from LV14 (results/grading_chart.png), at TMS 0.225 for R4.

At the maximum faults all four relays grade with at least 0.369 s between them (rule 0.30 s). At the minimum fault at the end of a string, R4 at TMS 0.225 would trip at 4.43 s before R3 at 6.19 s: the curves cross because the pickups sit at different multiples of the fault current. With the minimum-fault check in the pass rule (D3), R4 goes to TMS 0.35: R4 then takes 6.89 s at that fault (0.70 s after R3), and 3.9 s for a maximum fault at the offshore 66 kV busbar against 1.55 s before. A 3.9 s backup time is long; whether it is acceptable depends on the transformer's through-fault withstand, which is not assessed. A real scheme would add unit protection (differential on the transformers and cables) so that overcurrent is backup only.

R1 (D7): with the final loads the string relay's pickup must be at least 1.2 x 794.1 A = 952.9 A, and at most 0.8 x its zone-end minimum fault (6925 A) = 5540 A. At 960 A it trips in 0.2402 s at the string-feeder fault, 0.4435 s ahead of R2. Fault currents move by under 0.3 percent on revision D, so no other grading verdict changes (FAULT-D).

9. N-1

Five single outages, each in the four cases. Verdicts by the rule in section 4. 'Curtailed' gives the largest total output inside limits; 'outside limits even at zero output' means curtailing cannot cure it (a voltage set by the grid or by cable charging, not by the farm's output).

Outage	Case	Verdict	Worst exceptions at full output	Export (A)	Max trafo %	PCC (pu)
O1	C1	CURTAILED: 246.1 MW	loading EXP2 152.16	1135.0	97.4	0.9751
O1	C2	OUTSIDE LIMITS even at zero output	voltage PCC 0.9397; loading EXP2 165.65	1235.6	86.7	0.9397
O1	C3	CURTAILED: 256.2 MW	voltage ON220 0.8852; voltage OSS220 0.8783	1316.2	120.4	1.0066
O1	C4	RIDES THROUGH	none	421.2	31.5	1.0352
O2	C1	CURTAILED: 248.4 MW	loading T_OSS2 170.77	580.8	170.8	0.9837
O2	C2	OUTSIDE LIMITS	voltage PCC	670.9	165.4	0.9488

		even at zero output	0.9488; loading T_OSS2 165.43			
O2	C3	CURTAILED: 235.2 MW	voltage ON220 0.8845; voltage OSS220 0.8756	750.8	228.9	1.0064
O2	C4	RIDES THROUGH	none	397.8	18.2	1.0481
O3	C1	CURTAILED: 239.8 MW	loading T_ON2 178.08	619.9	178.1	0.9831
O3	C2	OUTSIDE LIMITS even at zero output	voltage PCC 0.9484; loading T_ON2 165.84	707.8	165.8	0.9484
O3	C3	CURTAILED: 216.4 MW	no solution	-	-	-
O3	C4	RIDES THROUGH	none	395.9	17.5	1.0481
O4	C1	RIDES THROUGH	none	527.6	66.2	0.9933
O4	C2	RIDES THROUGH	none	620.3	67.0	0.9539
O4	C3	RIDES THROUGH	none	454.4	73.8	1.0319
O4	C4	RIDES THROUGH	none	389.6	8.1	1.0476
O5	C1	RIDES THROUGH	none	598.0	78.6	1.0077
O5	C2	RIDES THROUGH	none	701.3	87.7	0.9702
O5	C3	RIDES THROUGH	none	545.0	83.1	1.0437
O5	C4	VOLTAGE EXCEPTION (not curable by curtailment)	voltage PCC 1.0651; voltage ON220 1.1068	422.2	30.5	1.0651

O1 one export circuit out; O2 one offshore transformer out; O3 one onshore transformer out; O4 the first section of string 1 out (string 1 lost); O5 the onshore reactor out. Source: results/n1.json.

What the table says

- Export circuit, offshore transformer or onshore transformer out: the farm must curtail to about 240 to 250 MW at unity power factor, and to 216.4 to 256.2 MW in C3, where the farm also absorbs reactive power. The limit is the remaining 250 MVA transformer or the remaining cable's rating: two half-rated items cannot carry the whole output.
- In C2 (grid at 0.95 pu, its lower limit) the same three outages leave the PCC just under 0.95 pu even at zero output (0.9397, 0.9488, 0.9484 pu at full output). That voltage is set by the grid; it needs reactive support or tap changing, which this model does not have (taps fixed, A19).
- One onshore transformer out in C3 does not solve at full output: the load flow does not converge, a sign that one transformer cannot carry 400 MW while absorbing reactive power at a high grid voltage. Curtailed, it holds 216.4 MW.
- A string's first section out loses only that string's 80 MW and rides through in every case.
- One reactor unit out rides through at full output in C1 to C3. At light load (C4) the PCC reaches 1.0651 pu and internal buses 1.1299 pu, which curtailment cannot cure: the remedy would be tap changing or switching the reactor in steps. In revision C, with one 310 MVAr reactor, losing it put every case out of limits at full output (internal voltages up to 1.2024 pu).

10. From revision C to revision D

Revision C was run first, by its own criteria (criteria_C.json). It showed three things, each tried first as a labelled what-if: the array cable had to be iterated (W3: 1000 mm2 clears C3), the single reactor was a single point of failure (W4: two 155 MVAr units, one out, keeps C1 to C3 inside limits), and R1's pickup sat 0.5 percent under its own rule. Revision D takes all three into the base design (D5 to D7), with criteria written before it was run (criteria_D.json). Revision C stays in the record (results/n1.json, checks.json) so the step can be traced.

Not run, and named as the next steps a real study would take: a third export circuit or a single larger HVAC or HVDC link if N-1 at full output were required; tap-changing transformers; turbine reactive capability; reactor switching in steps for light load; a cost comparison of losses against cable size (LV14's W2 showed the minimum-size export cable roughly doubles losses).

11. Checks and verification

Check	Result	What it shows
SOLVE (rev D)	FAIL	Every run converges. One does not: one onshore transformer out at C3, full output. Counted as outside limits; reported as a finding.
BAL (rev D)	PASS	Generation minus power into the grid equals loss; worst 4.1e-09 MW.
HAND-O2 (rev D)	PASS	Curtailed output with one offshore transformer out, 248.4 MW, against 250 MW by hand (0.64 %).
FAULT-D	PASS	Largest change in any fault level on the final cables 0.266 %, under the 2 % threshold.
R1 (D7)	PASS	Pickup rule met (952.9 to 5540 A) and 0.4435 s to R2.
HAND-O1 (rev C)	FAIL	Expected the remaining export circuit to carry the intact total within 5 %: 1134.7 A against 2 x 605.3 A (6.27 %). The rule was wrong: half the charging current leaves with the circuit that is out. Reported as failed, not adjusted.
LV14 independent	PASS	Export cable rating (759.6 A) and the 66 kV fault level (13.23 kA) re-derived without the LV14 code: twelve values within 0.12 %.
Tests	all pass	tests/test_lv15.py on revisions C and D from a clean shell; LV12 20, LV13 6 and LV14 9 also rerun.

Limits of the checking: the independent checks use the same general-knowledge formulas as the model, so they check the arithmetic and the model build, not the standards. No second engineer has reviewed the study.

12. What this study is not

- Not a design for construction, and not professional experience. Every input is assumed.
- Not a compliance assessment: the Grid Code, the SQSS (including its offshore chapter) and the IEC standards named were not read.
- Intact and single-outage steady state only: no dynamics, no fault ride-through, no harmonics or resonance (long cables and reactors make these real questions), no earth faults, no switching transients.
- Turbines are fixed P and Q injections: no turbine transformer, no reactive capability curve, no control.
- No economics: cable size against losses, a third circuit against curtailment, and reactor units against risk are named but not costed.

13. How to reproduce

With Python 3 and the packages in requirements.txt (pandapower 3.5.6): in LV12, casefiles/fetch_data.py then pytest; in LV13, pytest; then LV14/connection-study/run_all.sh and LV15/connection-study/run_all.sh (revisions C and D). Each block's results are rewritten and its tests rerun. This report is built by model/report.py from the saved results.

14. Sources

No standard, grid code, datasheet or survey was read for this study; nothing is cited as read. Named for the method only: IEC 60287 (cable rating), IEC 60228 (conductor sizes), IEC 60909 (short-circuit currents), IEC 60255-151 (relay curves). pandapower 3.5.6 (open source) was used and its short-circuit source code read. Every number in this report comes from a results file of LV13, LV14 or LV15, named beside it.

Appendix A. Study notes: what to be able to explain without notes

Ten ideas this study rests on, each with the number from this study that shows it. They are written for the author's own exam on the study; each answer should take under two minutes at a whiteboard.

Why long AC cables need reactors. A cable is a long capacitor. One 60 km, 220 kV circuit of the 630 mm2 cable takes about 131 MVAr to energise (V squared x omega x C x length). Two circuits make about 263 MVAr that must go somewhere; without a reactor it lifts every voltage (LV13: 1.14 pu internally) and loads the onshore transformers.

Why the cable rating depends on the soil. The rating is the current whose heat the ground can carry away for a 75 K rise. Soil from 1.0 to 1.5 K m/W cuts the 500 mm2 export rating from 760 A to 667 A, which is why revision C sizes on 1.5 until a survey says otherwise.

Why a bigger cable can still be overloaded. A bigger conductor also has more capacitance, so more charging current. The 800 mm2 array cable carries 796 A in C3 against its 790 A rating; sizing must be iterated until it settles (1000 mm2).

The voltage-source fault method. Put a voltage source c x Un / root 3 at the fault, short every other source, and divide by the impedance seen from the fault. c is 1.10 for the maximum (to cover operating voltage above nominal) and 1.00 for the minimum. At the 66 kV offshore bus: 1.1 x 0.8748 kA / 0.07276 pu = 13.23 kA from the grid alone.

Why transformers get a correction factor. K_T = 0.95 x cmax / (1 + 0.6 xT) scales transformer impedance in the maximum case; here 0.964. It makes the maximum fault current a little larger, so it errs on the safe side for equipment ratings.

Maximum and minimum faults do different jobs. The maximum fault sizes equipment (breaking capacity, cable short-circuit withstand) and sets the grading margins. The minimum fault checks that relays still see the fault and still trip in the right order; it is what caught R3 and R4 crossing.

The standard inverse curve and grading. t = TMS x 0.14 / ((I/Is)^0.02 - 1). Pickup above the load (1.2 x) and below the minimum fault (0.8 x); the relay nearer the fault must trip at least 0.30 s before the one behind it, at every fault level, not only the largest.

What N-1 means here. Take out one element at a time and ask whether the rest stays inside limits. Two half-rated transformers or cables cannot carry full output when one is out, so the farm curtails to about 250 MW. Whether that is acceptable is a planning-standard and cost question, not a load-flow one.

Why the reactor is split in two. Losing a single 310 MVAr reactor (revision C) took internal voltages to 1.12 to 1.20 pu, and curtailing the farm does not help because the charging comes from the cable, not the turbines. With two 155 MVAr units (revision D), one out rides through at full output; only light load stays over voltage, which needs taps or reactor switching.

What makes the study honest. Criteria committed before the first run; every number from code that reruns; a second method or hand check for each result; every failure reported (one run that does not converge, one hand check whose rule was wrong); and every input marked ASSUMED.

Appendix B. Bus voltages, revision D, intact network

Bus	C1 (pu)	C2 (pu)	C3 (pu)	C4 (pu)
PCC	0.9902	0.9537	1.0246	1.0481
ON220	0.9627	0.9643	0.9533	1.0421
OSS220	0.9797	0.9917	0.9582	1.0557

OSS66	0.9810	1.0298	0.9176	1.0631
S1T1	0.9826	1.0328	0.9176	1.0635
S1T2	0.9832	1.0340	0.9175	1.0636
S1T3	0.9838	1.0350	0.9175	1.0638
S1T4	0.9842	1.0359	0.9175	1.0639
S1T5	0.9846	1.0366	0.9175	1.0639
S1T6	0.9849	1.0371	0.9175	1.0640
S1T7	0.9850	1.0375	0.9175	1.0640
S1T8	0.9851	1.0377	0.9175	1.0641
S2T1	0.9826	1.0328	0.9176	1.0635
S2T2	0.9832	1.0340	0.9175	1.0636
S2T3	0.9838	1.0350	0.9175	1.0638
S2T4	0.9842	1.0359	0.9175	1.0639
S2T5	0.9846	1.0366	0.9175	1.0639
S2T6	0.9849	1.0371	0.9175	1.0640
S2T7	0.9850	1.0375	0.9175	1.0640
S2T8	0.9851	1.0377	0.9175	1.0641
S3T1	0.9826	1.0328	0.9176	1.0635
S3T2	0.9832	1.0340	0.9175	1.0636
S3T3	0.9838	1.0350	0.9175	1.0638
S3T4	0.9842	1.0359	0.9175	1.0639
S3T5	0.9846	1.0366	0.9175	1.0639
S3T6	0.9849	1.0371	0.9175	1.0640
S3T7	0.9850	1.0375	0.9175	1.0640
S3T8	0.9851	1.0377	0.9175	1.0641
S4T1	0.9826	1.0328	0.9176	1.0635
S4T2	0.9832	1.0340	0.9175	1.0636
S4T3	0.9838	1.0350	0.9175	1.0638
S4T4	0.9842	1.0359	0.9175	1.0639
S4T5	0.9846	1.0366	0.9175	1.0639
S4T6	0.9849	1.0371	0.9175	1.0640
S4T7	0.9850	1.0375	0.9175	1.0640
S4T8	0.9851	1.0377	0.9175	1.0641
S5T1	0.9826	1.0328	0.9176	1.0635
S5T2	0.9832	1.0340	0.9175	1.0636
S5T3	0.9838	1.0350	0.9175	1.0638
S5T4	0.9842	1.0359	0.9175	1.0639
S5T5	0.9846	1.0366	0.9175	1.0639
S5T6	0.9849	1.0371	0.9175	1.0640
S5T7	0.9850	1.0375	0.9175	1.0640
S5T8	0.9851	1.0377	0.9175	1.0641

Appendix C. Decision log

Date	Decision	By
18 Oct 2026	LV13 criteria revision A written before the first run	Claude Code, from Jack's brief
21 Oct 2026	Decision 7: build on revision B, 310 MVAr reactor in every case	Jack
22 Oct 2026	LV14 criteria written before the first run; cables sized at soil 1.0 K m/W	Claude Code
27 Oct 2026	D1 to D4: size on soil 1.5; revision C; minimum-fault check in the pass rule, R4 TMS 0.35; all inputs ASSUMED	Proposed by Claude; approved by Jack 28 Oct
27 Oct 2026	LV15 criteria revision C written before the first run (commit 34caf6a)	Claude
28 Oct 2026	D5 to D7: array 1000 mm2; reactor as two 155 MVAr units; R1 960 A. Criteria revision D written before its run (commit df09ba4)	Claude, on Jack's instruction to finalise with the improvements